NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science and Engineering
MID SEMESTER EXAMINATION, Jan-June 2024

Course Code: **CS64120**
Course Name: **Data Mining**

Max. Marks: 30
Maximum Time: 2 hours

Instructions:

1. Attempt all questions.
2. Assume any suitable data, if necessary. *Give diagram where necessary.*
3. Answer all the questions in the order as appeared in the question paper and write all the sub-parts of a question in one place.

S.N.	Questions	Marks	CO	BL
1.	a) In the process of data mining, how do we handle missing values and noisy data?	6	CO1	Remember,
	b) Data for price of items are as follows: 13, 13, 10, 12, 12, 17, 17, 15, 33, 35, 36, 17, 17, 18, 22. Use bin size as 5, smooth this price data by ***bin means*** and ***bin boundaries.***	4	CO1	Apply
2.	a) What do you mean by the OLAP? What are the various OLAP operations in multidimensional data model?	6	CO2	Remember, Analyze,
	b) Suppose the TAJ sales contain the following dimensions: *city, item, and year.* Draw the Lattice of cuboids for TAJ sales.	4	CO2	Apply
3.	Consider the given transactional dataset and answer the following: a) Compute the support for itemsets $\{e\}$, $\{b,d\}$, $\{a,d\}$, and $\{b,d,e\}$ by treating each transaction ID as a market basket.	2	CO3	Apply
	b) Use the results in part (a) to compute the confidence for the association rules $\{b,d\} \rightarrow \{e\}$ and $\{e\} \rightarrow \{b,d\}$. Is confidence a symmetric measure? Justify.	2	CO3	Apply
	c) Apply the Apriori algorithm to extract all frequent itemsets by considering minsup = 30%. Demonstrate each step.	6	CO3	Apply

Customer ID	Transaction ID	Items Bought
1	0001	$\{a,d,e\}$
1	0024	$\{a,b,c,e\}$
2	0012	$\{a,b,d,e\}$
2	0031	$\{a,c,d,e\}$
3	0015	$\{b,c,e\}$
3	0022	$\{b,d,e\}$
4	0029	$\{c,d\}$
4	0040	$\{a,b,c\}$
5	0033	$\{a,d,e\}$
5	0038	$\{a,b,e\}$

e e e
b c d
a a a
a b e